

Solução do Problema BRANDY - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: BRANDY
- Número de Variáveis: 249
- Número de Restrições: 220
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: 1.519e+03
- Valor Dual: 1.519e+03
- Gap: 4.929e-20
- Número de Iterações: 17

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema BRANDY

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	7.360e+01	14	2.866e-02
2	1.445e+03	20	8.377e-01
3	8.100e+00	23	9.218e-02
4	3.438e+01	24	9.608e-02
5	5.962e+01	25	1.323e-01
6	2.000e-01	26	8.222e-03
7	1.664e+01	28	5.131e-02
9	8.000e+00	29	1.048e-02
10	1.400e+01	32	7.004e-01
11	2.375e+01	33	8.576e-02
12	3.256e+01	34	2.824e-02
13	5.000e+00	35	1.341e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
15	2.244e+01	36	5.211e-02
16	2.600e+00	38	7.039e-03
18	3.000e+00	39	6.233e-03
19	6.000e-01	40	2.017e-01
21	2.195e+00	41	3.240e-02
22	2.000e+00	43	2.271e-01
27	1.808e+01	44	1.801e-01
30	2.979e+01	45	5.772e-01
31	4.341e+00	46	4.472e-01
37	4.666e-02	47	3.728e-02
42	2.915e+01	48	6.794e-02
50	9.119e+00	49	4.461e-01
51	2.290e+01	57	2.202e-01
52	9.288e+00	58	8.175e-02
53	2.858e+00	68	3.226e-02
54	3.713e+00	70	7.046e-01
55	4.598e+00	71	5.563e-02
56	2.455e+01	72	1.721e-02
59	5.226e+01	74	5.563e-02
60	8.408e+00	75	1.721e-02
63	1.321e+00	77	8.333e-02
64	9.958e+00	78	3.743e-02
66	2.237e+01	83	4.580e-03
69	6.608e+00	91	1.556e+00
76	1.155e+01	106	7.162e-02
79	8.647e+00	117	2.510e-01
80	1.670e+01	118	2.016e-01
81	7.260e+00	120	4.121e-01
82	6.135e+00	121	1.905e-02
84	1.300e+01	124	1.190e-02
85	8.035e-01	126	9.424e-03
86	4.000e+00	127	3.263e-02
87	5.394e+00	128	4.995e+01
88	4.627e+00	129	5.004e+01
89	1.060e+00	133	2.584e-02
90	7.822e+00	134	7.713e-02
96	4.625e+00	135	5.063e-02
97	2.073e-01	139	1.905e-02
98	1.604e-01	142	1.190e-02
99	1.908e-02	143	3.554e-03
100	1.622e-01	146	4.994e+01
101	5.414e+00	147	4.997e+01
102	1.089e+01	153	1.937e-01
103	9.748e-01	154	9.670e-02
104	3.309e+00	156	7.736e-02
105	5.107e+00	159	3.519e-01
107	7.232e-01	160	1.104e+00
109	6.297e+00	162	1.805e-01
111	3.211e+00	165	1.046e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
112	1.379e+00	170	6.715e-03
113	3.022e+00	173	1.086e-01
115	5.465e+00	174	1.358e-02
116	7.692e+00	177	5.002e+01
119	6.938e+00	184	3.521e-01
122	9.562e-01	189	5.869e-01
123	6.464e+01	190	1.090e+00
125	6.560e+01	194	6.686e-01
130	2.367e+00	195	4.976e+01
131	4.235e+00	199	4.394e-02
132	3.110e-01	201	3.795e+00
136	8.385e+00	204	3.419e-01
137	5.924e+00	209	1.451e+00
138	1.780e-01	226	1.303e+00
140	1.493e+01	227	5.144e+00
141	6.466e+00	228	4.196e+01
144	7.561e+00	229	4.271e+01
145	1.384e+01	230	4.419e+01
148	2.000e+00	232	9.197e+00
149	1.700e+00	233	8.039e+00
150	1.697e+00	234	7.292e+00
151	4.003e+00	235	5.815e+00
152	3.300e+00	245	5.815e+00
155	1.030e+01	246	5.660e+00
157	9.000e+00	247	5.684e+00
158	3.000e+00	248	6.314e+00
161	5.868e-02	249	5.933e+00
163	4.911e+00		
164	1.604e-01		
166	6.629e+00		
167	6.537e+00		
168	4.421e+00		
169	5.076e+00		
171	4.610e+00		
172	5.958e-01		
175	3.300e+01		
176	3.300e+01		
178	1.065e+03		
179	3.849e+00		
180	5.011e-01		
181	6.500e-01		
182	4.000e+00		
183	1.340e+01		
185	8.670e+00		
186	1.146e+00		
187	2.244e+00		
188	4.229e+00		
191	6.120e-01		
192	1.690e+01		

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
193	1.432e+00		
196	3.038e+00		
197	7.449e+00		
198	4.940e+00		
200	2.600e+00		
202	9.304e-01		
203	1.584e+00		
205	3.040e+00		
206	2.153e+01		
207	1.527e+00		
208	9.029e-01		
210	1.570e+00		
211	7.200e+00		
212	1.329e+03		
217	2.100e+00		
218	5.000e-01		
219	2.900e+00		
220	2.500e+00		
221	7.250e-01		
222	1.426e+02		
223	4.745e-01		
224	8.018e-01		
225	1.442e+03		
231	1.000e-01		

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores $\lambda \geq 0$ do problema BRANDY

Coordenada y	Valor y (Dual)
25	9.856e+00
26	9.197e+00
27	1.154e+01
28	8.433e+00
29	8.303e+00
30	8.039e+00
31	7.959e+00
32	7.292e+00
33	6.140e+00
34	6.185e+00
35	5.843e+00
36	6.491e+00
37	5.815e+00
38	8.252e+00
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Coordenada y	Valor y (Dual)
40	4.754e-01
41	1.093e+00
42	6.166e+00
43	7.717e+00
44	7.659e+00
45	7.658e+00
46	8.079e+00
47	7.992e+00
48	8.103e+00
49	8.202e+00
50	8.202e+00
51	8.202e+00
52	7.461e+00
53	6.674e+00
54	6.147e+00
56	5.574e+00
57	6.995e+00
60	8.202e+00
61	8.294e+00
62	8.105e+00
63	7.562e+00
64	7.430e+00
65	7.417e+00
67	6.662e+00
69	8.887e+00
70	9.293e+00
72	9.311e+00
74	9.368e+00
75	9.468e+00
77	9.453e+00
79	9.472e+00
80	8.122e+00
81	9.813e+00
82	4.669e+00
83	5.684e+00
84	6.307e+00
85	7.293e+00
86	6.770e+00
89	5.660e+00
90	6.186e+00
91	6.024e+00
92	6.166e+00
94	7.780e+00
95	6.696e+00
96	6.362e+00
97	7.925e+00
99	8.104e+00
100	7.153e+00
101	6.262e+00

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Coordenada y	Valor y (Dual)
103	2.321e+01
104	1.786e+01
105	9.622e+00
106	9.767e+00
107	9.895e+00
109	2.000e+01
115	7.659e+00
116	7.658e+00
117	7.659e+00
118	7.658e+00
119	1.033e+01
120	9.497e+00
122	9.411e+00
127	4.747e-01
128	1.046e+01
134	8.081e+00
137	6.426e+00
139	6.380e-03
140	5.815e+00
141	6.166e+00
144	8.542e+00
147	7.157e+00
148	1.084e-02
150	8.024e+00
151	8.230e+00
152	5.741e+00
153	7.659e+00
155	9.444e+00
157	9.501e+00
193	6.139e+00
195	5.860e-03
200	5.574e+00
201	6.790e+00
202	6.864e+00
203	3.827e+00
208	5.896e+00
209	6.799e+00